Yuhang Li

Ph.D. student, The University of Tokyo · trustworthy machine learning

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RESEARCH

I work on trustworthy machine learning: learning and reasoning from unreliable information. Real-world labels are partial or wrong, classes are imbalanced, and inputs can be manipulated. I build models that can tell reliable from unreliable information, and evaluation methods that verify whether they do.

- **Weakly supervised and robust learning.** Learning from labels that are incomplete, noisy, or long-tailed. *Now:* re-examining gains in long-tailed learning against baselines with tuned post-hoc adjustment.
- **Reliable evaluation.** Measuring model behavior, including uncertainty, against a well-defined ground truth. *Now:* grading models against an ideal learner in synthetic worlds where the true probability of every outcome is known.
- **Safety and privacy of foundation models.** How models can be manipulated through their inputs and memory, and how to protect them. *Now:* whether one perturbed image in a multimodal assistant's context can store a behavior, not only an answer.

EDUCATION

The University of Tokyo Oct 2026 – present
Ph.D. student, Complexity Science and Engineering, Graduate School of Frontier Sciences
Advisor: Prof. Masashi Sugiyama.

Southeast University Sep 2023 – Jun 2026
M.Eng., Computer Science and Technology
Advisor: Prof. Yuheng Jia, PAttern Learning and Mining (PALM) Lab.

Dalian University of Technology Sep 2019 – Jun 2023
B.Eng., Computer Science and Technology
Advisor: Prof. Xin Yang.

PUBLICATIONS

Complementary Label Learning with Positive Label Guessing and Negative Label Enhancement ICLR 2025
Yuhang Li, Zhuying Li, Yuheng Jia paper · code
Treats complementary-label learning as an inverse problem: guesses a likely positive label and enhances the negative labels to recover a richer training signal. 55.25% accuracy on STL-10 with one complementary label per example, +20.29 points over POCR (34.96%).

Boosting Class Representation via Semantically Related Instances for Robust Long-Tailed Learning with Noisy Labels ICCV 2025
Yuhang Li, Zhuying Li, Yuheng Jia paper · code
Strengthens class representations by transferring semantic soft labels from related instances and combining shot-specific experts for head, medium, and tail classes. 69.80% top-1 accuracy on CIFAR-10 with 50% label noise and imbalance ratio 200, +4.99 points over SFA (64.81%).

Samples Are Not Equal: A Sample Selection Approach for Deep Clustering ICLR 2026
Zhengxing Jiao, Yaxin Hou, Jun Ma, **Yuhang Li**, Ding Ding, Yuheng Jia, Hui Liu, Junhui Hou paper · code

Robust Long-Tailed Partial Label Learning by Head Classifier and Tail Classifier Cooperation under review
Y. Jia, **Y. Li**, X. Peng, R. Wang, M. Zhang manuscript
First student author. A head classifier and a tail classifier cooperate, each handling the regime it is good at, so that label ambiguity and class imbalance do not amplify each other.

RESEARCH EXPERIENCE

Huawei 2012 Laboratories Jun – Oct 2026

Research Intern, Theory Research Department

Mentors: Dr. Wei Zhang and Dr. Hongming Huang.

- Built request-arrival-triggered prefix KV-cache prefetching in vLLM + LMCache for long-context inference. The SSD-to-host transfer starts when a request arrives, so I/O overlaps with the time the request spends in the queue.
- Measured +31.57% request throughput and −24.49% mean time to first token over native prefetch in a 3,379-request replay (12-layer validation model, one output token per request). A related patent disclosure passed internal review.
- Implemented and validated asynchronous prefix prefetching in vLLM-Ascend + Unified Cache Management (UCM): cache residency protection, shared I/O and multi-rank coordination, cancellation and fallback, on both MLA and standard GQA paths.

RIKEN Center for Advanced Intelligence Project (AIP) Dec 2025 – May 2026

Student Trainee, Imperfect Information Learning Team

Supervisors: Dr. Gang Niu and Prof. Masashi Sugiyama.

- Weakly supervised learning with foundation models under long-tailed distributions and noisy labels.

HONORS

2027	JSPS Research Fellowship for Young Scientists (DC1), tenure from April 2027
2026	BOOST NAIS, The University of Tokyo (JST-supported)
2025	National Scholarship for Graduates, China
2025	SI-TECH “Future Star” Education Fund, Second Prize Scholarship

SERVICE AND SKILLS

Reviewer	NeurIPS 2026, ICLR 2026, IEEE MLMC 2025/2026; TMLR, Pattern Recognition, IJMLC
Research	Python, PyTorch; weak supervision, robust and long-tailed learning
Systems	vLLM, LMCache, vLLM-Ascend, UCM; KV-cache prefetching; Linux, Git, Slurm
Languages	Mandarin (native), English (IELTS 7.5)